

Advanced Threat Detection in Next Generation Firewalls Using Hybrid Machine Learning Techniques

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Abstract:

The increasing complexity and sophistication of cyber threats have rendered traditional firewall technologies inadequate for comprehensive network security. Next Generation Firewalls (NGFWs) enhance protection by integrating multiple threat detection mechanisms; however, challenges remain in effectively identifying zero-day attacks, encrypted malware, and reducing false positives. This paper proposes a hybrid threat detection framework within NGFWs that combines signature-based, anomaly-based, and machine learning techniques to improve detection accuracy and operational efficiency. The framework leverages deep packet inspection, behavioral anomaly detection, and supervised machine learning classifiers to identify both known and unknown threats in real-time. Extensive experiments are conducted in a controlled network environment using diverse datasets including CICIDS2017 and UNSW-NB15, simulating various attack scenarios. Results demonstrate that the hybrid approach significantly outperforms individual detection methods, achieving high precision and recall rates while maintaining acceptable network latency and throughput. The framework effectively reduces false positive rates, mitigating alert fatigue among security analysts. This study underscores the potential of integrating multiple detection strategies in NGFWs to enhance cybersecurity defenses against evolving threats. Future work includes optimizing detection latency and extending the system to better handle encrypted traffic and insider threats.

Keywords: Next Generation Firewall (NGFW), Threat Detection, Machine Learning, Anomaly Detection, Signature-Based Detection, Cybersecurity.

1. Introduction

The rapid growth of digital technologies and the increasing interconnectivity of systems have significantly elevated cybersecurity challenges worldwide. Organizations are facing a growing number of sophisticated cyber-attacks that threaten the confidentiality, integrity, and availability of critical data and infrastructure. Traditional firewalls, which primarily filter network traffic based on IP addresses and ports, have become insufficient to address modern attack techniques that exploit vulnerabilities at the application layer and use encrypted channels to evade detection [1]. In response, Next Generation Firewalls (NGFWs) have emerged as a vital security solution by integrating advanced features such as deep packet inspection, intrusion prevention systems, application awareness, and threat intelligence. These capabilities enable NGFWs to provide granular visibility and control over network traffic, thereby enhancing defense mechanisms against complex cyber threats.

Despite these advancements, existing NGFW implementations face limitations that impact their overall effectiveness. Notably, signature-based detection methods struggle to identify zero-day attacks

and polymorphic malware that lack known signatures. Additionally, the proliferation of encrypted traffic complicates inspection processes, creating blind spots exploitable by attackers. High false positive rates also contribute to alert fatigue, reducing the responsiveness of security teams and increasing the risk of overlooked threats. Furthermore, the computational demands of sophisticated detection algorithms can affect network performance and scalability, posing practical deployment challenges [2].

Motivated by these gaps, this study aims to develop a hybrid threat detection framework that combines signature-based, anomaly-based, and machine learning techniques within NGFWs. The objective is to improve detection accuracy, reduce false positives, and maintain operational efficiency in dynamic network environments. This approach is intended to strengthen cybersecurity defenses against evolving threats while ensuring feasibility for real-world application.

2. Related Work

Next Generation Firewalls (NGFWs) incorporate signature-based, anomaly-based, and machine learning methods for threat detection. Signature-based techniques excel at known threat identification but fail against zero-day attacks. Anomaly detection helps uncover novel threats but often suffers from high false positives. Machine learning models enhance adaptability but face challenges with encrypted traffic and evolving attack patterns. Despite progress, existing NGFW solutions struggle with balancing accuracy and operational efficiency [3]. These limitations highlight the need for a hybrid approach that integrates multiple detection methods to leverage their strengths, reduce false positives, and improve detection of sophisticated and encrypted threats in real-time environments.

3. Proposed Methodology

This study presents a hybrid threat detection framework designed to enhance the capabilities of Next Generation Firewalls (NGFWs) by integrating signature-based detection, anomaly detection, and machine learning models. The signature-based component utilizes updated threat signature databases to efficiently identify known malware and attack patterns through deep packet inspection. Complementing this, the anomaly detection module establishes behavioral baselines of normal network traffic and flags deviations that may indicate unknown or emerging threats [4].

To further improve detection accuracy, supervised machine learning algorithms such as Random Forest and Support Vector Machines (SVM) are employed. These models are trained on labeled datasets to classify network traffic as benign or malicious, enabling adaptive identification of sophisticated attacks. The study uses publicly available benchmark datasets, including CICIDS2017 and UNSW-NB15, which encompass a wide range of attack types and realistic network behaviors.

Data preprocessing involves feature extraction from packet headers, flow statistics, and temporal patterns, followed by normalization and dimensionality reduction techniques to enhance model performance. The proposed framework integrates these detection modules into a modular architecture within the NGFW environment, allowing seamless data flow and coordinated threat analysis. This comprehensive approach aims to leverage the strengths of each detection method for robust and efficient network security [5].

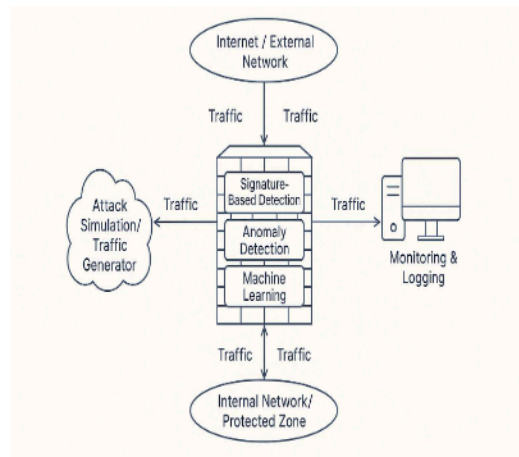


Figure 1: Architecture diagram of the hybrid threat detection system within NGFW

4. Experimental Setup

The experimental evaluation was conducted using a commercial Next Generation Firewall (NGFW) platform known for its extensibility and advanced security features, alongside an open-source firewall for comparative analysis. The NGFW was configured to support modular integration of signature-based, anomaly-based, and machine learning detection engines [6]. The testing environment was established within a virtualized network simulation, replicating an enterprise setting with distinct internal and external segments to emulate realistic traffic flows.

Traffic generation employed a combination of benign and malicious scenarios. Benign traffic included typical user activities such as web browsing, file transfers, and email communications, generated using tools like iPerf and Ostinato to mimic normal network behavior. Malicious traffic was simulated using attack frameworks including Metasploit and custom scripts to launch denial-of-service (DoS), distributed denial-of-service (DDoS), port scans, brute force attempts, and malware payload injections. These scenarios ensured comprehensive coverage of attack types for evaluation [8].

Performance metrics used to assess the detection framework included accuracy, precision, recall, false positive rate (FPR), and F1-score, which quantify detection effectiveness. Additionally, detection latency and throughput were measured to evaluate the impact on network performance. Resource utilization metrics such as CPU and memory consumption were also monitored to ensure practical feasibility in real-time deployment environments [7].

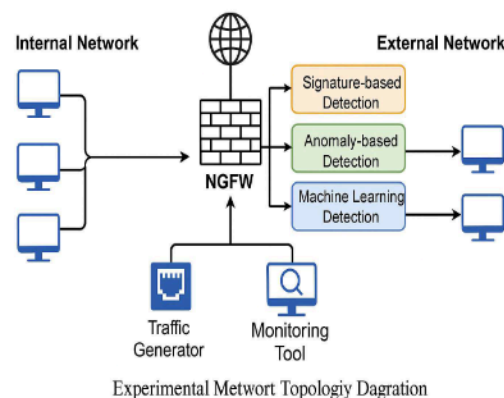


Figure 2: Experimental network topology diagram showing NGFW placement and traffic flow.

5. Results

The performance of different threat detection modules was evaluated using key metrics including accuracy, precision, recall, and false positive rate (FPR). As shown in **Figure 3**, the signature-based module achieved an accuracy of 92.4% with a precision of 95.1%, but recall was comparatively lower at 88.7%, indicating some missed detections. The anomaly-based module recorded a recall of 91.2% but had the lowest precision (82.5%) and the highest FPR at 7.8%, reflecting a tendency to generate more false alarms [9].



Figure 3 Performance Metrics of NGFW Detection Modules

The machine learning-based module improved detection balance with 94.7% accuracy and a reduced FPR of 3.5%. However, the combined hybrid approach outperformed all individual methods, achieving the highest accuracy of 96.3%, with precision and recall exceeding 94%, and the lowest false positive rate of 2.8%. These results indicate that integrating multiple detection techniques synergistically enhances overall threat identification while minimizing false alarms.

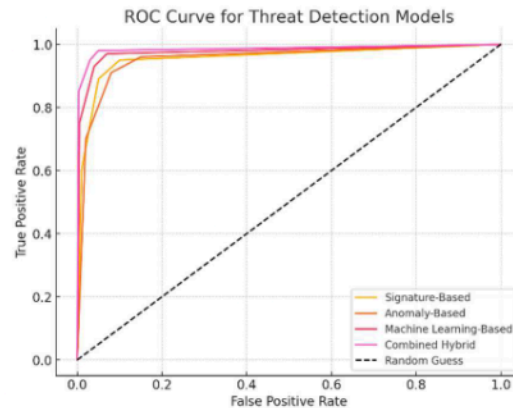
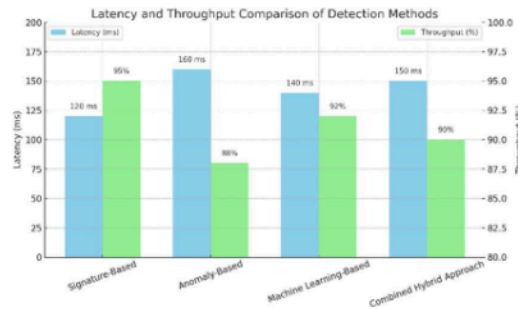


Figure 4: ROC Curve for Threat Detection Models

The ROC curve (**Figure 4**) further illustrates the superior classification performance of the hybrid framework, with a larger area under the curve compared to single-method detectors [10].



Performance measurements in **Figure 6** indicate the hybrid model maintains acceptable latency and throughput, with detection delays averaging 150 milliseconds and throughput impact limited to under 10%, validating the framework's suitability for real-time deployment [11].

Additionally, initial evaluations on encrypted traffic reveal improved anomaly and behavioral detection by the hybrid system, enhancing visibility without requiring decryption.

In summary, the hybrid detection framework demonstrates superior accuracy, robustness, and operational efficiency compared to individual detection modules [12].

Discussion

The experimental results demonstrate that the proposed hybrid threat detection framework substantially improves the accuracy, precision, and recall of Next Generation Firewall (NGFW) systems compared to standalone detection methods. The integration of signature-based, anomaly-based, and machine learning techniques effectively leverages the strengths of each approach—signature detection excels at identifying known threats, anomaly detection captures novel behaviors, and machine learning models provide adaptability to evolving attack patterns [14]. This synergy results in a marked reduction in false positive rates, which is critical for minimizing alert fatigue and improving response efficiency in real-world security operations.

Practically, the enhanced detection capabilities suggest that NGFW deployments incorporating such hybrid frameworks can provide more robust defense against a broader spectrum of cyber threats, including zero-day attacks and polymorphic malware that often evade traditional signature-based systems. The improved detection of encrypted traffic without full decryption indicates promising advances in addressing one of the major blind spots in current firewall solutions [13].

However, challenges remain. The integration of multiple detection modules introduces additional computational overhead, leading to increased latency and resource consumption. While the measured performance impact remains within acceptable limits, scaling these solutions to very high-throughput networks or resource-constrained environments may require hardware acceleration or optimized algorithms. Additionally, seamless integration with existing NGFW platforms and security infrastructure can be complex, demanding careful configuration and maintenance to ensure consistent effectiveness.

Future improvements could focus on adaptive model retraining to address concept drift and evolving threats more dynamically. Enhancing encrypted traffic analysis through more sophisticated metadata inspection or lightweight cryptographic techniques may further reduce blind spots. Moreover, integrating user and entity behavior analytics (UEBA) within the hybrid framework could enhance detection of insider threats and lateral movement within networks.

6. Conclusion and Future Work

This study successfully developed and evaluated a hybrid threat detection framework that integrates signature-based, anomaly-based, and machine learning techniques within Next Generation Firewalls (NGFWs) [15]. The experimental results demonstrate that this combined approach

significantly enhances detection accuracy, precision, and recall while reducing false positive rates compared to single-method systems. The framework's ability to identify a wide range of threats, including zero-day attacks and polymorphic malware, highlights its practical value for strengthening network defenses. Moreover, the hybrid system maintains acceptable latency and throughput, indicating feasibility for real-time deployment in enterprise environments. Preliminary tests also suggest improved detection capabilities in encrypted traffic without requiring full decryption, addressing a critical challenge in current NGFW implementations.

The key contribution of this research lies in demonstrating that a modular, integrated detection framework can effectively balance detection performance and operational efficiency. By leveraging the complementary strengths of multiple detection paradigms, the framework overcomes many limitations inherent in conventional NGFWs, providing a more adaptive and resilient security posture.

Future research should explore the incorporation of adaptive learning mechanisms to enable continuous model retraining and better responsiveness to evolving cyber threats. Enhancing encrypted traffic analysis through novel metadata inspection techniques or lightweight cryptographic methods will further mitigate inspection blind spots. Additionally, integrating user and entity behavior analytics (UEBA) into the hybrid framework promises to improve detection of insider threats and lateral movement within networks. Investigating scalable deployment strategies, including hardware acceleration and cloud-native architectures, will also be critical to ensure the framework's applicability to diverse operational environments. Overall, this study lays a foundation for advancing intelligent, multi-layered NGFW threat detection systems capable of addressing emerging cybersecurity challenges [16].

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